

Red Hat Data Grid 8.6

Data Grid Code Tutorials

Learn how to use Data Grid capabilities

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Abstract

Run code tutorials for remote caches and embedded caches that demonstrate various Data Grid capabilities and usage patterns.

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RED HAT DATA GRID

Data Grid is a high-performance, distributed in-memory data store.

Schemaless data structure

Flexibility to store different objects as key-value pairs.

Grid-based data storage

Designed to distribute and replicate data across clusters.

Elastic scaling

Dynamically adjust the number of nodes to meet demand without service disruption.

Data interoperability

Store, retrieve, and query data in the grid from different endpoints.

DATA GRID DOCUMENTATION

Documentation for Data Grid is available on the Red Hat customer portal.

- Data Grid 8.6 Documentation
- Data Grid 8.6 Component Details
- Supported Configurations for Data Grid 8.6
- Data Grid 8 Feature Support
- Data Grid Deprecated Features and Functionality

DATA GRID DOWNLOADS

Access the Data Grid Software Downloads on the Red Hat customer portal.



NOTE

You must have a Red Hat account to access and download Data Grid software.

MAKING OPEN SOURCE MORE INCLUSIVE

Red Hat is committed to replacing problematic language in our code, documentation, and web properties. We are beginning with these four terms: master, slave, blacklist, and whitelist. Because of the enormity of this endeavor, these changes will be implemented gradually over several upcoming releases. For more details, see our CTO Chris Wright's message.

CHAPTER 1. REMOTE CACHES

Deploy multiple Data Grid Server instances to create remote cache clusters that give you a fault-tolerant and scalable data tier with high-speed access from Hot Rod and REST clients.

1.1. REMOTE CACHE TUTORIALS

To run these tutorials you need at least one locally running instance of Data Grid Server. Each tutorial will try to connect to a running server in **localhost:11222** with **admin**/**password** credentials. However, if a Docker instance is found, and the server is not running, tutorials will spin up a local server with **Testcontainers**.

You can download the distribution and run the following commands:

```
$ ./bin/cli.sh user create admin -p "password"
$ ./bin/server.sh
```



NOTE

Data Grid Server enables authentication and authorization by default. Creating a user named **admin** gives you administrative access to Data Grid Server.

Building and running remote cache tutorials

You can build and run remote cache tutorials directly in your IDE or from the command line as follows:

```
$ ./mvnw -s /path/to/maven-settings.xml clean package exec:exec
```

1.2. HOT ROD JAVA CLIENT TUTORIALS

- Data Grid requires Java 11 at a minimum. However, Hot Rod Java clients running in applications that require Java 8 can continue using older versions of client libraries.

Tutorial link	Description
Remote cache use example	The simplest code example that demonstrates how a remote distributed cache works.
Remote cache with cache alias example	Demonstrates how to update a cache config to add an alias and use it to manipulate the cache.
Per cache configuration	Demonstrates how to configure caches dynamically when we connect to the Data Grid Server.
Near caches	Demonstrates how configure near caching to improve the read performance in remote caches.
Cache Admin API	Demonstrates how to use the Administration API to create caches and cache templates dynamically.

Tutorial link	Description
Encoding	Demonstrates how encoding of caches work.
Custom marshalling	Demonstrates how custom marshalling works in the context of programmatically creating Protostream schemas and marshallers. Users should generally rely on annotations based generated Protostream marshallers unless exceptional circumstances require manual creation.
Protostream schemas administration	Demonstrates how to use the Administration API to manipulate the Protostream (protobuf format) schemas in Data Grid.
Client listeners	Detect when data changes in a remote cache with Client Listeners.
Query	Demonstrates how to query remote cache values.
Continuous query	Demonstrates how to use Continuous Query and remote caches.
Transactions	Demonstrates how remote transactions work.
Secured caches	Demonstrates how to configure caches that have authorization enabled.
TLS authorization	Demonstrates how to connect to Data Grid Server with TLS authorization.
Multimap	Demonstrates how remote multimap works.
Task execution	Demonstrates how to register server tasks and how to execute them from the Hot Rod client.
JUnit 5 and Testcontainers	Demonstrates how to use the Data Grid and JUnit 5 extension.
Persistence	Demonstrates how to use the Data Grid and persistent caches.
Redis Client	Demonstrates how to use the Data Grid and Redis client to read and write using the Resp protocol.
Reactive API	Demonstrates how to use the Data Grid with the reactive API based on Mutiny.

Data Grid documentation

You can find more resources for Hot Rod Java clients in our documentation at:

- Hot Rod Java client guide
- Marshalling and Encoding Data Guide
- Querying Data Grid caches
- REST API
- Resp Protocol
- Smallrye Mutiny

CHAPTER 2. EMBEDDED CACHES

Add Data Grid as a dependency to your Java project and use embedded caches that increase application performance and give you capabilities to handle complex use cases.

2.1. EMBEDDED CACHE TUTORIALS

You can run embedded cache tutorials directly in your IDE or from the command line as follows:

```
$ ./mvnw -s /path/to/maven-settings.xml clean package exec:exec
```

Tutorial link	Description
Distributed caches	Demonstrates how Distributed Caches work.
Replicated caches	Demonstrates how Replicated Caches work.
Invalidated caches	Demonstrates how Invalidated Caches work.
Cache alias	Demonstrates how Cache alias work.
Transactions	Demonstrates how transactions work.
Cache security	Demonstrates how to integrate an application with Data Grid security features.
Streams	Demonstrates how Distributed Streams work.
JCache integration	Demonstrates how JCache works.
Functional Maps	Demonstrates how Functional Map API works.
Map API	Demonstrates how the Map API works with Data Grid caches.
Multimap	Demonstrates how to use Multimap.
Metrics	Demonstrates how to enable and extract metrics in Data Grid.
Queries	Uses Data Grid Query to perform full-text queries on cache values.
Query with programmatic index definition	Uses Data Grid Query to perform full-text queries on cache values, with programmatic index definition.
Clustered Listeners	Detects when data changes in an embedded cache with Clustered Listeners.

Tutorial link	Description
Counters	Demonstrates how to use an embedded Clustered Counter.
Clustered Locks	Demonstrates how to use an embedded Clustered Lock.
Clustered execution	Demonstrates how to use an embedded Clustered Counter.

Data Grid documentation

You can find more resources about embedded caches in our documentation at:

- Embedding Data Grid Caches
- Querying Data Grid caches

2.2. KUBERNETES AND OPENSHIFT TUTORIAL

This tutorial contains instructions on how to run Infinispan library mode (as a microservice) in Kubernetes/OpenShift.

Prerequisites: Maven and Docker daemon running in the background.

Prerequisites

- A running Openshift or Kubernetes cluster

Building the tutorial

This tutorial is built using the maven command:

```
./mvnw package
```

Note that **target**/ directory contains additional directories like **docker** (with generated Dockerfile) and **classes**/**META-INF**/**jkube** with Kubernetes and OpenShift deployment templates.

TIP

If the Docker Daemon is down, the build will omit processing Dockerfiles. Use **docker** profile to turn it on manually.

Deploying the tutorial to Kubernetes

This is handle by the JKube maven plugin, just invoke:

```
mvn k8s:build k8s:push k8s:resource k8s:apply -Doptions.image=<IMAGE_NAME> (1)
```

(1) **IMAGE_NAME** must be replaced with the FQN of the container to deploy to Kubernetes. This container must be created in a repository that you have permissions to push to and is accessible from within your Kubernetes cluster.

Viewing and scaling up

Everything should be up and running at this point. Now login into the OpenShift or Kubernetes cluster and scale the application

```
kubectl scale --replicas=3  deployment/$(kubectl get rs --namespace=myproject | grep infinispan | awk '{print $1}') --namespace=myproject
```

Undeploying the tutorial

This is handled by the JKube maven plugin, just invoke:

```
mvn k8s:undeploy
```

CHAPTER 3. SPRING AND SPRING BOOT

3.1. SPRING AND SPRING BOOT TUTORIALS



NOTE

These code tutorials use Data Grid Server and require at least one running instance.

Run Spring examples

Two simple tutorials can be run with Spring without Spring Boot:

- Test caching

```
$ ./mvnw -s /path/to/maven-settings.xml package exec:exec@spring-caching
```

- Test annotations

```
$ ./mvnw -s /path/to/maven-settings.xml package exec:exec@spring-annotations
```

To run the Spring Boot tutorials, use the following command.

```
$ ./mvnw -s /path/to/maven-settings.xml spring-boot:run
```

Displaying actuator statistics

Navigate to **http://localhost:8080/actuator/metrics** in your browser to display a list of available metrics. Cache metrics are prefixed with **"cache."** Display each metric for each cache using tags. For example for the 'puts' stats in the basque-names cache:

http://localhost:8080/actuator/metrics/cache.puts?tag=name:basque-names

Collecting statistics with Prometheus

The **prometheus.yml** file in this project contains a *host.docker.internal* binding that allows Prometheus to scrap metrics that the Spring actuator exposes.

Change the **YOUR_PATH** value in the following command to the directory where Prometheus is running and then run:

Podman

```
$ podman run -d --name=prometheus -p 9090:9090 -v YOUR_PATH/integrations/spring-
boot/prometheus.yml:/etc/prometheus/prometheus.yml prom/prometheus --
config.file=/etc/prometheus/prometheus.yml
```

Tutorial link	Description
Spring Boot and Spring Cache remote mode	Demonstrates how to use Spring Caches with Spring Boot and the Data Grid Server.

Tutorial link	Description
Spring Boot, Spring Cache, Reactor and Infinispan Server	Demonstrates how to use Spring Caches with Spring Boot Reactor and the Data Grid Server.
Spring Boot and Spring Session remote mode	Demonstrates how to use Spring Session with Spring Boot and the Data Grid Server.
Spring Boot and Spring Cache embedded mode	Demonstrates how to use Spring Caches with Spring Boot and Data Grid Embedded.
Spring Boot and Spring Session embedded mode	Demonstrates how to use Spring Session with Spring Boot and Data Grid Embedded.
Spring cache embedded without Spring Boot	Demonstrates how to use Spring Cache and Data Grid Embedded without Spring Boot.

Data Grid documentation

You can find more resources in our documentation at:

- Using Data Grid with Spring
- Data Grid Spring Boot Starter